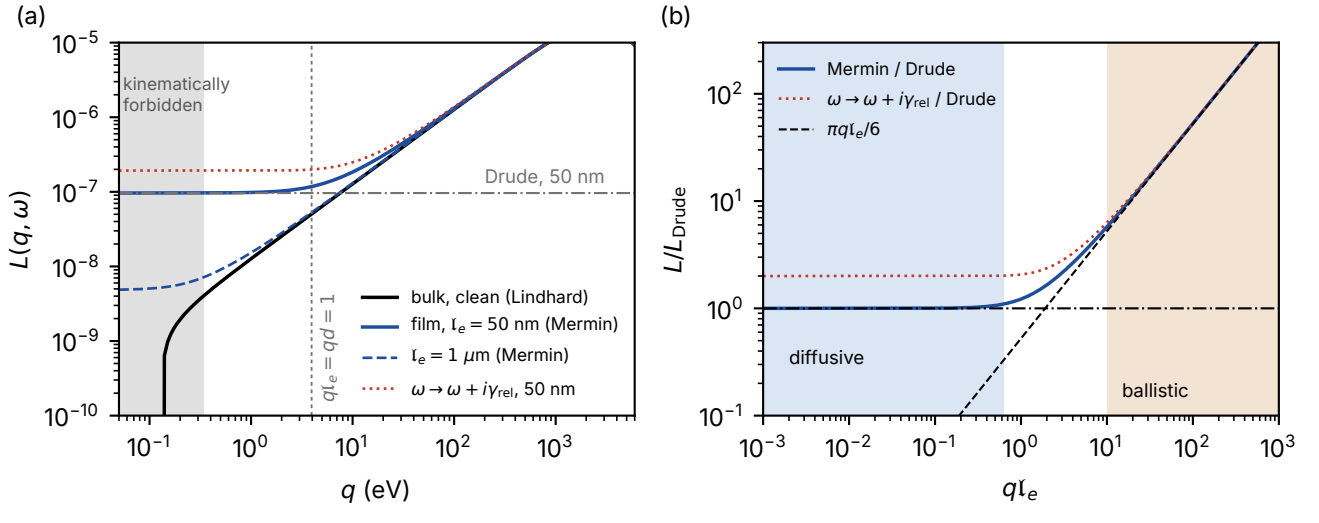
(a)
kinematically forbidden
Drude, 50 nm
$q\mathfrak{l}_e = qd = 1$
bulk, clean (Lindhard)
film, $\mathfrak{l}_e = 50$ nm (Mermin)
$\mathfrak{l}_e = 1$ μm (Mermin)
$\omega \to \omega + i\gamma_{\mathrm{rel}}$, 50 nm
$L(q, \omega)$
q (eV)
(b)
Mermin / Drude
$\omega \to \omega + i\gamma_{\mathrm{rel}}$ / Drude
$\pi q \mathfrak{l}_e/6$
diffusive
ballistic
L/L_{Drude}
$q\mathfrak{l}_e$